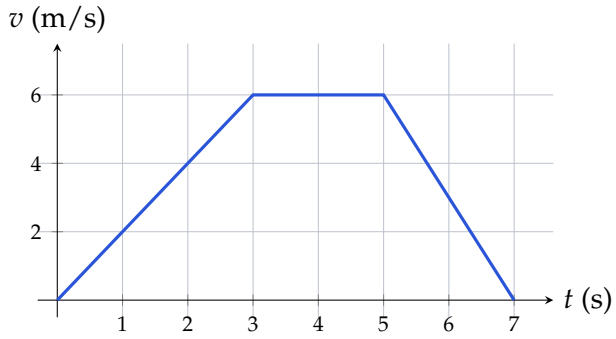
AP PHYSICS 1

Unit 1 • Lesson 1.3 • Guided notes

Topic 1.3 • Representing Motion

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: read and draw position, velocity and acceleration graphs, and move between them.

Position vs. time	On an x–t graph, the _____ is the velocity. Straight line: _____. Curved line: the velocity _____ (acceleration). Horizontal line: the object is _____.
Velocity vs. time	On a v–t graph: the _____ is the acceleration; the _____ between the graph and the t-axis is the displacement. Area below the axis counts as _____ displacement. Where the graph crosses the axis, the object _____.
Acceleration vs. time	On an a–t graph, the _____ is the change in velocity Δv.
Example: a v–t graph	 Accelerations: 0–3 s: _____ 3–5 s: ____ 5–7 s: _____ Displacement = total area = _____ Average velocity over 7 s: _____
Motion diagrams	Dots show the position at equal time intervals. equal spacing: _____ spacing grows: _____
Summary	